higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MASONRY
NQF LEVEL 3

11 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1: GENERAL

1.1 False
1.2 False
1.3 True
1.4 True
1.5 False
1.6 False
1.7 True
1.8 False
1.9 False
1.10 False

(10 × 1) **[10]**

QUESTION 2: CAVITY WALL CONSTRUCTION

2.1 2.1.1 DPC is placed in a wall.
2.1.2 DPM is placed under the concrete slab.

(2 × 2) (4)

2.2
- Butterfly wall ties
- Plastic wall ties
- Double-triangle wall ties
- PWD wall ties
- Single-wire wall ties

(Any 4 × 1) (4)

2.3 Weep holes are created in a cavity wall to drain moisture from the cavity. (2)

2.4
- Bottom of wall above ground level
- Below window sill
- Above window openings
- Above door
- Below doorframe
- At door and window jambs

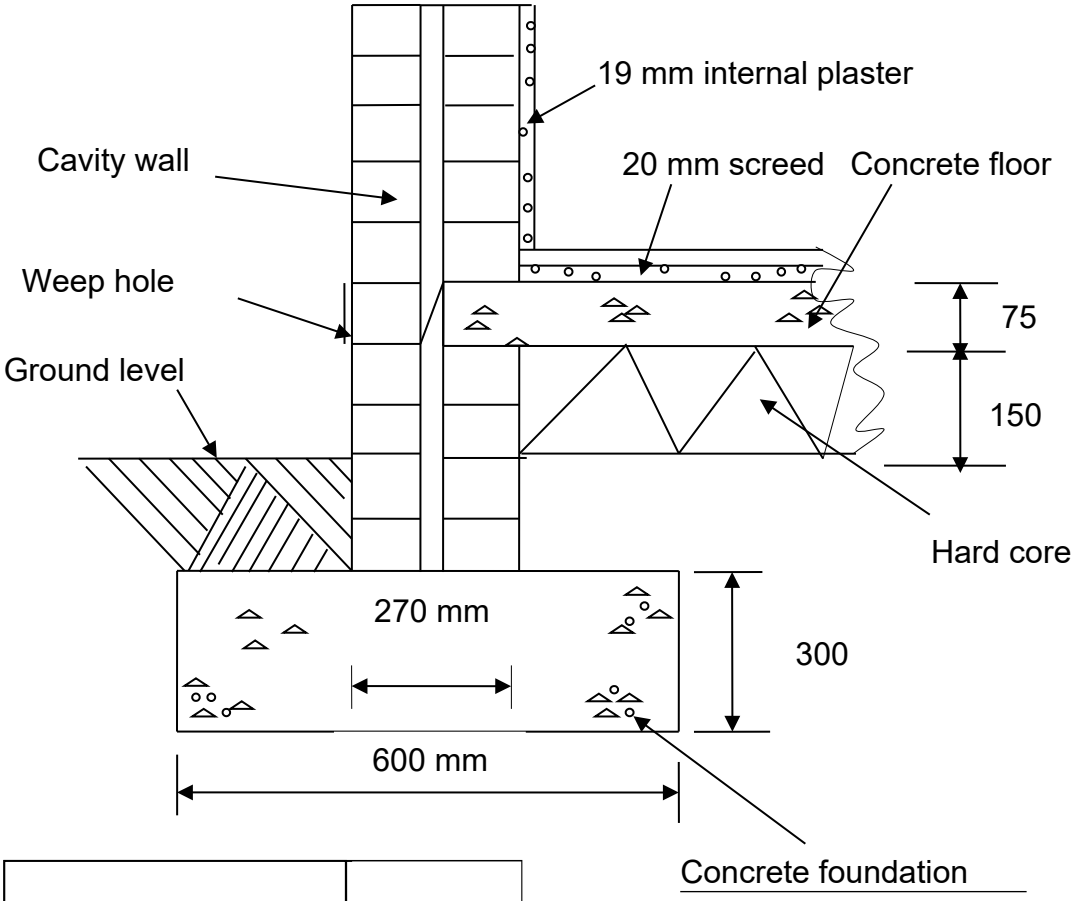
(6)

2.5
- Set in the concrete infill above the lintel at an angle.
- Lay the DPC on the concrete infill.
- Tuck the DPC into the shell bedding on the cavity face.
- Make a V-joint in the plaster on the outer leaf at the level of the DPC.
- Make weep holes in the vertical joint above the lintel.

(5)

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2.6



Correctness	2
Dimensions	1
Labelling	6

(9)
[30]

QUESTION 3: BASIC MASONRY WALL CONSTRUCTION WITH HOLLOW UNITS

3.1
- Portland cement
- Water
- Sand
- Gravel (Any 3 × 1) (3)

3.2
- Gauge and mark profiles.
- Measure and correct foundation footing.
- Check variance of mortar bed.
- Mark out base of wall.
- Lay units on shell beddings. (Any 4 × 1) (4)

3.3 Place a 3 m long straight edge on two masonry units of the exact size.✓ Take a second straight edge of the same length and place one end on one of the masonry units next to the first straight edge✓ and leave the other end free to move up and down.✓ Place a spirit level on the free end✓ and lift the straight edge or lower it until it is level. Measure the distance between the top of the first straight edge✓ and the top of the second straight edge.✓ If the distance is bigger than 6 mm the foundation should be levelled✓ before brickwork can commence.✓ If the first check is correct measure one metre from first check✓ and check level✓ and continue until whole foundation is checked.✓ (10)

3.4
- Round jointer and pointing trowel
- V-jointer and pointing trowel
- Pointing trowel and mortar trowel
- Bricklaying trowel
- Square jointer and scraper
- Pointing trowel and mastic trowel (Any 5 × 1) (5)

3.5
- Full unit - most of the walls
- ¾ unit - wall
- ½ unit - intersections, returns
- ¼ unit - doorframes, window frames and lintels
- Sill unit - window sills
- Lintel unit - lintels
- Sash unit - window frames
- Bond unit - lintels/ring beams (Any 4 × 2) (8)

[30]

QUESTION 4: ADVANCED MASONRY WALL CONSTRUCTION

4.1
- Lightweight roof – 600 mm
- Heavyweight roof – 300 mm (2)

4.2 4.2.1 Voussoirs are the individual wedge-shaped bricks in an arch.

4.2.2 The span is the distance between the jambs or reveals of the opening of an arch.

4.2.3 The keystone/key brick is the highest or central brick and usually the last to be built in.

4.2.4 Extrados is the upper edge of the arch as seen from the front.

4.2.5 The crown is the highest point of an arch.

4.2.6 Intrados is the underside edge of an arch.

(6 × 2) (12)

4.3 •Galvanised steel wire (hurricane brackets)
•Metal strap/hoop iron (2)

4.4 Not less than seven days after completion of the installation (2)

4.5 You will be held accountable for any injury and be forced by the law to pay any damages. (2)

4.6 •If you have to use acid, dilute hydrochloric acid (spirit of salts) with water or a special masonry cleaner.
•Always soak the masonry with clean water before you apply the acid dilution.
•Use the diluted acid to clean the mortar droppings and smears off the brickwork.
•Rinse the brickwork thoroughly so that no acid remains on it.
•If necessary, use chemical cleaning agents, but follow the manufacturer's instructions to not damage the masonry. (5)

4.7 •Measuring tape
•Spirit level
•Trowel
•Bolster (for cutting bricks)
•Hummer (for cutting bricks)
• Steel square (Any 5 × 1) (5)
[30]

TOTAL: 100

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